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Arnett Rule

- radiation dominated $\Rightarrow E(\text{erg g}^{-1}) = aT^4V$; $V = \rho^{-1}$
- coasting, homologous expansion

$$\Rightarrow R(\text{photosphere}) = R_0 + tu$$

$$\Rightarrow \frac{\dot{V}}{V} = 3 \frac{\dot{u}}{u}$$

1st law of thermo :

$$\dot{E} + P\dot{V} = \epsilon - \frac{\partial u}{\partial m} \quad \begin{matrix} \nearrow \text{cooling luminosity} \\ \downarrow \text{energy input per gram} \end{matrix}$$

In the one-zone approximation :

$$\frac{\partial u}{\partial m} \sim \frac{L}{m} \sim \frac{E}{T_d}$$

T_d = radiative diffusion time ;

$$R^2 \sim D T_d \quad ; \quad D \sim \lambda c \quad ; \quad \lambda = (\text{MFP}) = \frac{1}{\rho \kappa}$$

$$\Rightarrow T_d \sim \frac{R^2}{\lambda c} = \frac{\rho \kappa R^2}{c} = \frac{\rho R^3 \kappa}{c R}$$

$$\sim \frac{M \kappa}{c R} \quad (\text{Arnett uses } \frac{M \kappa}{4 c R})$$

$$E = aT^4V \Rightarrow \frac{\dot{E}}{E} = 4\frac{\dot{T}}{T} + \frac{\dot{V}}{V}$$

$$\Rightarrow E(4\frac{\dot{T}}{T} + \frac{\dot{V}}{V}) + P \cdot V \cdot 3\frac{\dot{u}}{u} \approx E - \frac{L}{\tau_h}$$

$$= E(4\frac{\dot{T}}{T} + 3\frac{\dot{u}}{u}) + 3PV \cdot \frac{\dot{u}}{u}$$

$$P = \frac{1}{3}EV \Rightarrow E[4\frac{\dot{T}}{T} + 3\frac{\dot{u}}{u} + \frac{\dot{u}}{u}]$$

$$= E \cdot 4[\frac{\dot{T}}{T} + \frac{\dot{u}}{u}]$$

$$E(\text{ergs}^{-1}) \sim \frac{E}{\tau_h}; \quad \tau_h = \text{heating time}$$

$$\Rightarrow 4\left(\frac{\dot{T}}{T} + \frac{\dot{u}}{u}\right) = \frac{1}{\tau_h} - \frac{1}{\tau_c}$$

$\swarrow$ heating rate $\searrow$ cooling rate

For radioactive heating: $E = E_0 e^{-t/\tau_r}$

for ^{60}Ni , $E_0 = 4.8 \times 10^{10} \text{ ergs}^{-1}$, $\tau_r \sim 9 \text{ days}$

At very early times, when $\frac{u}{R} \gg \tau_c^{-1}, \tau_h^{-1}$, the evolution is nearly adiabatic

$$\Rightarrow \frac{\dot{T}}{T} \sim -\frac{\dot{R}}{R} \Rightarrow T \propto R^{-1}; \quad E \propto T^4 R^3 \propto R^{-1}.$$

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at times $\gg \tau_r$ (assuming no other heating), there is only radiative cooling:

$$\Rightarrow 4 \left(\frac{\dot{I}}{T} + \frac{u}{R} \right) \sim - \frac{4cR}{km} \quad (\text{as } R \text{ for } k \sim \text{const})$$

$$\Rightarrow 4 \frac{d \ln}{dt} (TR) \sim - \frac{4c}{km} \cdot \frac{d}{dt} \left(\frac{1}{2} u t^2 \right) \quad (\text{for } R \sim u t)$$

$$\frac{1}{2} \frac{R^2}{u}$$

$$\Rightarrow 4 d \ln (TR) = - \frac{4c}{km} \cdot \frac{1}{2u} \frac{d}{dt} R^2 \cdot dt$$

$$\Rightarrow 4 \ln (TR) \sim - \frac{4c}{km} \cdot \frac{1}{2u} \cdot R^2 + \text{const}$$

$$\Rightarrow \ln (TR)^4 \sim - \frac{1}{2} \cdot \frac{\tau_{exp}}{\tau_d} \rightarrow \text{expansion time}$$

$$\Rightarrow TR^4 \sim c e^{-\frac{1}{2} \tau_{exp} / \tau_d} = e^{-\frac{1}{2} \cdot R^2 / R_0^2}$$

$$R_0^2 = \left(\frac{4c}{km u} \right)^{-1} = \frac{km u}{4c}$$

$$L \sim \frac{4\pi R^2}{\tau_d} \sim \frac{4\pi}{km} 4cR \cdot T^4 R^3$$

$$= \frac{4c}{km} \cdot (TR)^4$$

$$\Rightarrow L \sim c e^{-R^2 / R_0^2} = c_1 \underline{\underline{e^{-c_2 t^2}}}$$

$\Rightarrow$ decaying luminosity.

⇒ must be an earlier time when LC peaks.

Since $L \propto (IR)^4$

$$\Rightarrow \frac{\dot{L}}{L} \sim 4 \left(\frac{\dot{I}}{I} + \frac{\dot{R}}{R} \right) = 4 \left(\frac{\dot{I}}{I} + \frac{4}{R} \right)$$

$$= \frac{1}{t_{\text{th}}} - \frac{1}{t_d}$$

$$= - \frac{4cR}{km} + \frac{E_0}{E} e^{-t/t_{\text{cr}}}$$

writing the internal energy as $E \sim E_0 t$
(appropriate for early times)

⇒ peak of LC @ $\dot{L} = 0$

$$\Rightarrow 0 = - \frac{4cR}{km} + t^{-1} e^{-t/t_{\text{cr}}}$$

In general, peak when rates of cooling + heating are equal

$$\Rightarrow t_{\text{peak}} \sim \frac{km}{4cR} \sim \frac{km}{4c u t_{\text{peak}}}$$

$$\Rightarrow t_{\text{peak}} \sim \left(\frac{km}{4cu} \right)^{1/2} = \left(\frac{R^2}{u} \right)$$

$$\sim \frac{1 \cdot 10^{33} \cdot 2}{4 \cdot 10^{10} \cdot 10^9 \cdot 3} \sim 10^{33-19} \sim 10''$$

$$\Rightarrow t_{\text{peak}} \sim \sqrt{10''} \approx 10^{5.5} \sim (5d)$$